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Wireless Power Transmission in a Closed Environment

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Rapid technological developments have driven the recent widespread interest in wireless power transmission technology, which has unique advantages for use in a variety of fields. Compared with traditional wired energy transmission, wireless power transmission does not require a physical connection, which provides greater flexibility and convenience, especially for situations that are challenging to wire or need a high degree of flexibility in application scenarios. In enclosed spaces, such as satellites, submarines, spacecraft, deep-sea probes, medical implantable devices, and future

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The number of devices in confined spaces is increasing, and the demand for energy supply is also growing.

smart home systems, the importance of wireless power transmission technology is self-evident [1], [2], [3], [4].

Enclosed space often has special physical properties and environmental constraints [5], such as space limitations, electromagnetic (EM) shielding, multiple receivers (Rxs), and weight limitations, which make wired energy transfer in these environments impractical or costly. Wireless power transmission technology overcomes these limitations and provides a stable and reliable power supply for electronic devices in enclosed spaces. In addition, with the continuous development of the Internet of Things (IoT), artificial intelligence, and other techniques, the number of devices in confined spaces is increasing, and the demand for energy supply is also growing, which makes research into wireless power transmission technology and its applications particularly important.

Enclosed space wireless power transmission technology involves several subject areas, including electromagnetism, microwave technology, signal processing, and control theory. Technically, it requires the realization of high-efficiency, high-stability, and high-security power transmission, and EM compatibility, energy loss, the multipath effect, and other issues must also be considered. Therefore, the study of enclosed space wireless power transmission technology has important academic value, a wide range of application prospects, and crucial strategic significance.

This article reviews the current status, principles, challenges, and future applications of enclosed space wireless power transmission technology. First, the basic principles and critical technologies of enclosed space wireless power transmission technology are introduced. Second, we analyze the applications and current problems of this technology in various fields. Finally, development trends and future research directions of this technology are discussed. It is hoped that the review in this article can provide readers with a comprehensive understanding and in-depth knowledge of enclosed space wireless power transmission technology.

Introduction

The concept of wireless power transmission dates back to the late 19th century, when it was demonstrated by Nikola Tesla's 1891 Wardencliff experiment [6]. Limited by the available technology, its development was slow until the mid-20th century, when electronic and microwave advancements led to gradually increasing research interest and significant breakthroughs. In 1968, Peter E. Glaser

proposed the solar satellite concept, emphasizing potential benefits and challenges [7]. W. C. Brown later verified microwave power transmission via helicopter experiments [8]. Since then, the research has been extended to space exploration and other fields, and the influence of the space environment on wireless power transmission systems has been analyzed. In 1974, Brown elaborated on the components and working principle of the free-space power transmission system [9]. In the 1990s, institutions, such as NASA, the Japanese government, and the U.S. Department of Defense, conducted experiments to explore the application of microwave power transmission in space on the ground and for military applications [10].

Early on, researchers explored the technology and application of free-space electrical energy transmission using microwave beams [9], which laid the foundation for wireless power transfer (WPT) technology. Magnetic coupling resonance theory later emerged as a key focus, with studies on transmission experiments, circuit theory, system analysis, and range adaptation [11], [12], [13], [14], all advancing short-to-medium-distance WPT applications. Design optimization, including numerical analysis for current control and efficiency, was also emphasized [15], [16], [17]. Recent breakthroughs in strongly coupled magnetic resonance have enabled mid-range WPT to multiple devices [18], [19], which further expands the application prospects of WPT technology.

With the maturity and increased application of WPT technology, converting microwave energy into dc efficiently is a key problem. The design and optimization of the rectifier circuits, a core component of a WPT system, are very important to improve efficiency. In recent years, researchers have designed a number of rectifier circuits to adapt to the application of WPT systems [20], [21], [22], [23], [24], including high efficiency, wide-input dynamic range, and broadband. High-efficiency rectifiers reduce energy loss, ensure sufficient electric energy for equipment in closed spaces, maintain system stability, and prolong equipment life. Rectifiers with a wide dynamic range adapt to changes in input power and improve system reliability. Broadband rectifiers enhance the compatibility of the system, making the closed-environment WPT system more suitable for complex applications.

With the maturity of WPT technology, there have come more and more application environments. In some specific environments, such as satellites, space stations, automobiles, and other similar enclosed air-spaces, enabling WPT has attracted the attention of scholars. Generally speaking, the existing methods for WPT in closed spaces can be grouped in two categories. The first method is to stimulate specific resonant modes in space and add energy-receiving units according to the field distribution of the resonant modes to realize power transfer. This power transfer method is based on the resonant cavity, and the

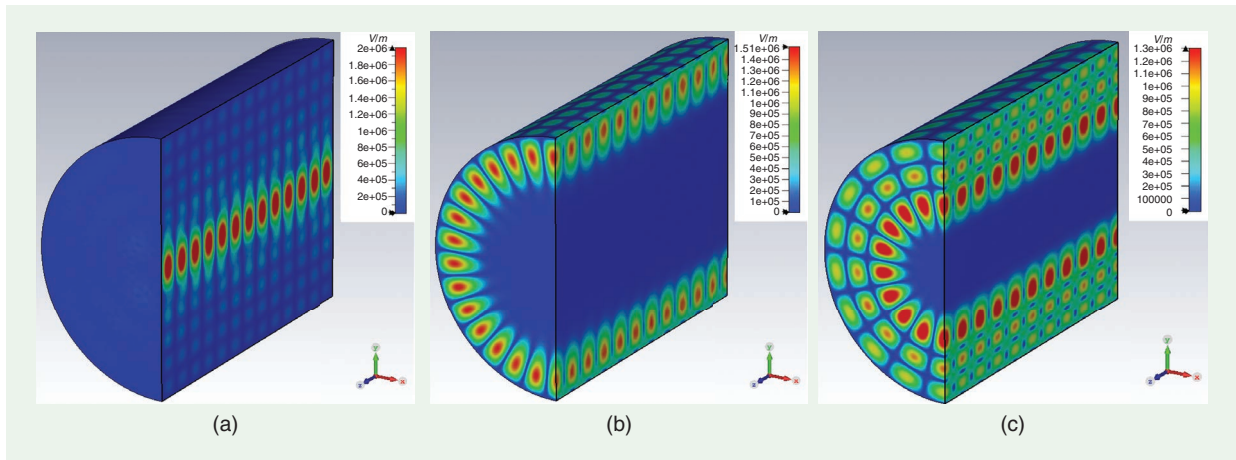


Figure 1. (a)–(c) Typical electric field distributions of a metal cylindrical cavity [25].

efficiency is high, but the energy transfer efficiency is greatly affected by the frequency and the location of the transceiver unit. The second method is based on multiple resonant modes of energy coupling transmission in a closed space to stimulate multiple resonant modes. Power transfer is made possible at most Rx unit locations by superimposing multiple modes of EM field distribution.

At its simplest, a microwave resonant cavity is a closed or semiclosed space. Standing waves will be formed when EM waves are transmitted, and their electric and magnetic fields are distributed according to certain rules. Taking a closed cylindrical metal cavity as an example, several typical electric field distributions are presented in Figure 1. There is a large number of standing waves in the resonant cavity. Adding a dipole antenna to the resonant cavity can realize energy transmission by coupling between the resonant cavity and the dipole antenna. Under varying resonant frequencies, the distribution of the excited electric field is nonuniform. By controlling a single frequency to select a suitable mode, the power can be concentrated in different positions inside the cavity. Based on the complementarity of the field distribution among modes, the power transmission of the receiving unit at any position in space can be realized by the coupling energy transfer method of multiresonant modes.

Development

The development of WPT technology in a closed environment is gradually opening a new chapter in energy utilization. With the rapid progress of science and technology and the growing demand for convenient and efficient energy, the limitations and inconveniences of traditional wired energy transmission methods are becoming more and more evident. In this context, WPT technology has become a hot spot for researchers to explore, especially for applications in closed environments. This technology

aims to realize safe and efficient energy transmission in specific closed spaces through noncontact methods, which is not only expected to enhance the flexibility and convenience of energy use greatly but also may bring revolutionary changes for medical devices, IoT sensors, smart homes, and even special environments, such as underwater and underground applications.

In 2011, Liou et al. proposed a broadband strongly coupled cavity resonator (CR) based on metamaterials, which utilizes artificial magnetic conductor (AMC) surfaces for

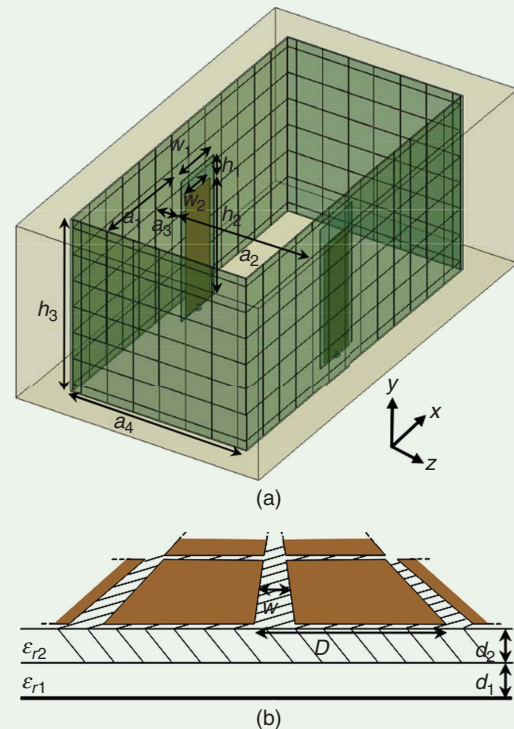


Figure 2. (a) The 3D geometry of a metamaterial-based cavity with two monopole-type antennas and (b) the AMC structure in [26].

The concept of wireless power transmission dates to the late 19th century, when it was demonstrated by Nikola Tesla's 1891 Wardencllyffe experiment.

efficient energy transfer [26]. The resonator achieves a uniform field distribution within the cavity and compact dimensions by combining four sidewall AMC surfaces and two top and bottom metal plates. AMC surfaces consist of a periodic arrangement of square patches printed on a metal-backed double-layer dielectric substrate, eliminating the need for through-hole connections and enhancing the operating bandwidth (Figure 2).

In 2013, Rahimzadeh et al. proposed a method to wirelessly power multiple electronic devices in an overmoded cavity [27]. An overmoded cavity is a cavity whose dimensions are greater than a certain multiple of the wavelength of an EM wave and is capable of forming complex EM field patterns inside the cavity. Through multimode excitation and narrow-band frequency modulation techniques, a relatively uniform EM field distribution is formed inside the overmoded cavity (Figure 3), which improves the efficiency and stability of the wireless power supply. The power management circuits integrated inside each receiving device can adaptively adjust the power distribution according to the battery level and the demand of the device, thus extending the endurance of the device.

In 2014, Korhummel et al. proposed an overmoded cavity-based wireless charging solution to enable

simultaneous, efficient, and uniform charging of multiple electronic devices [28]. A shielded overmoded conductive cavity was designed that was capable of exciting multiple EM modes at specific frequencies to create a relatively uniform EM field distribution inside the cavity. The transmitter (Tx) feeds energy into the overmoded cavity in an incoherent manner to excite the EM field inside the cavity. Due to the use of stirring methods, such as mechanical stirring, pseudorandom placement placement stirring, and frequency modulation stirring, multiple receiving devices can be placed anywhere inside the cavity (Figure 4) for charging without the need for precise alignment of the Tx. This design greatly improves the convenience and flexibility of wireless charging.

In 2015, Zajić and Popović presented a geometrical propagation model for a wireless powering channel in an overmoded cavity loaded with conductive objects [29]. A conductive resonator (Figure 5) approximately $50 \times 28 \times 18$ cm in size is excited by a 10-GHz probe, and the power distribution inside the charging box is measured using a rectenna and a power management circuit for various positions of other conductive objects within the resonator. Based on the geometrical model, a simulation model for multipath fading in this cavity is developed and compared with measured statistical data with 2,000 samples. The results show good agreement between the measured and simulated received power levels and distributions, implying that the presented model allows a statistical prediction that can be used for future designs of scalable overmoded cavities.

In 2015, Chabalko and Sample explored a key technical problem in the WPT field: how to realize the

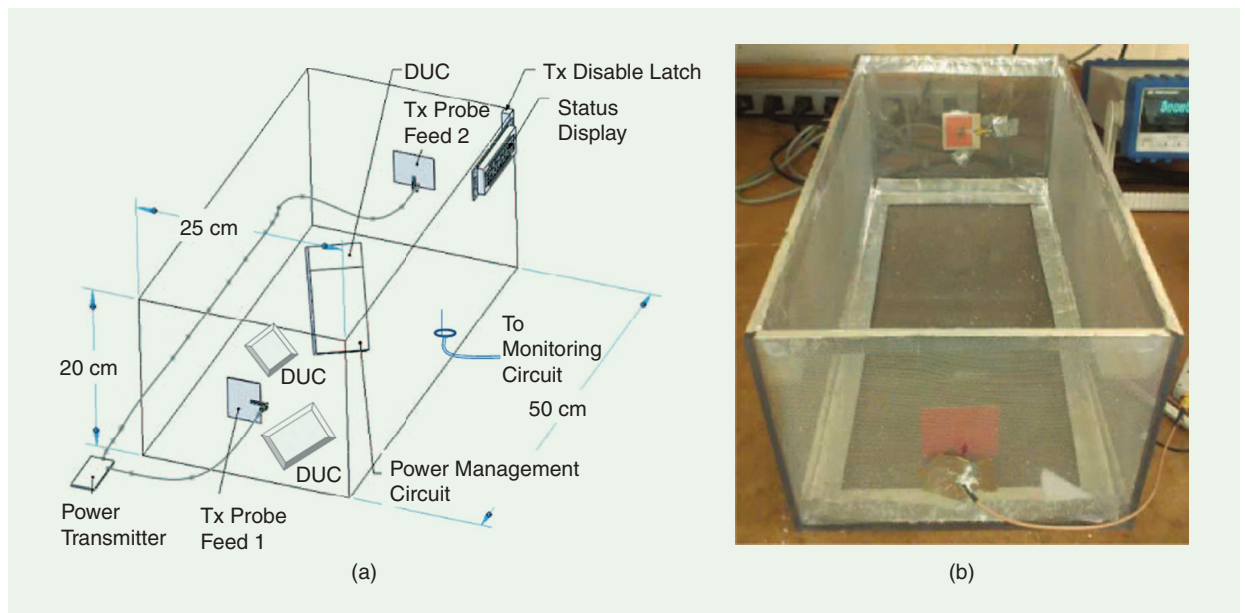


Figure 3. The (a) schematic diagram and (b) prototype of the overmoded rectangular cavity for wireless powering of multiple electronic devices in [27]. DUC: device under charge; Tx: transmitter.

effective coupling of electric fields to short dipole Rxs in a metallic cavity [30]. An expression for the coupling coefficient between the electric field of the resonant mode and a subwavelength resonant dipole Rx was derived based on coupled-mode theory (CMT). The results showed good agreement between analytically and numerically enabling a practical and solid prediction of the WPT efficiency.

In November of the same year, Chabalko and Sample presented a novel wireless power transmission system based on EM cavity resonance [31]. The system utilizes the natural EM patterns of a hollow metal structure to generate uniform magnetic fields that can simultaneously

power multiple small receiving coils located almost anywhere inside the structure. With two resonant modes, it was demonstrated that a 3-in-diameter Rx at virtually any location in a 140-ft² test chamber [Figure 6(a)] could be powered at more than 50% efficiency. The ability to power up to 10 Rxs [Figure 6(b)] simultaneously was also demonstrated, and the system could charge consumer electronic devices, such as cell phones. For the first time, a wireless power transmission system based on a multimode resonant cavity was demonstrated that has the capacity to charge multiple devices efficiently in three dimensions.

In 2016, Mei et al. proposed an impedance matching method for achieving optimal power transfer efficiency

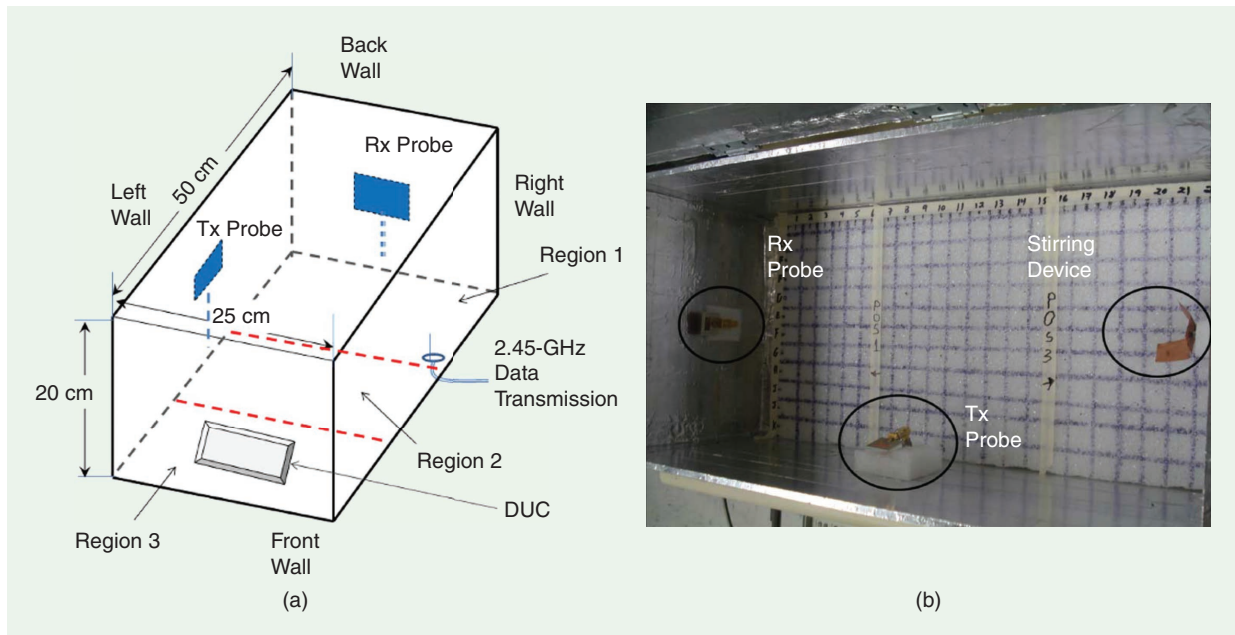


Figure 4. The (a) schematic diagram of the overmoded cavity used for powering multiple devices simultaneously and (b) cavity with a stirring mechanism in [28].

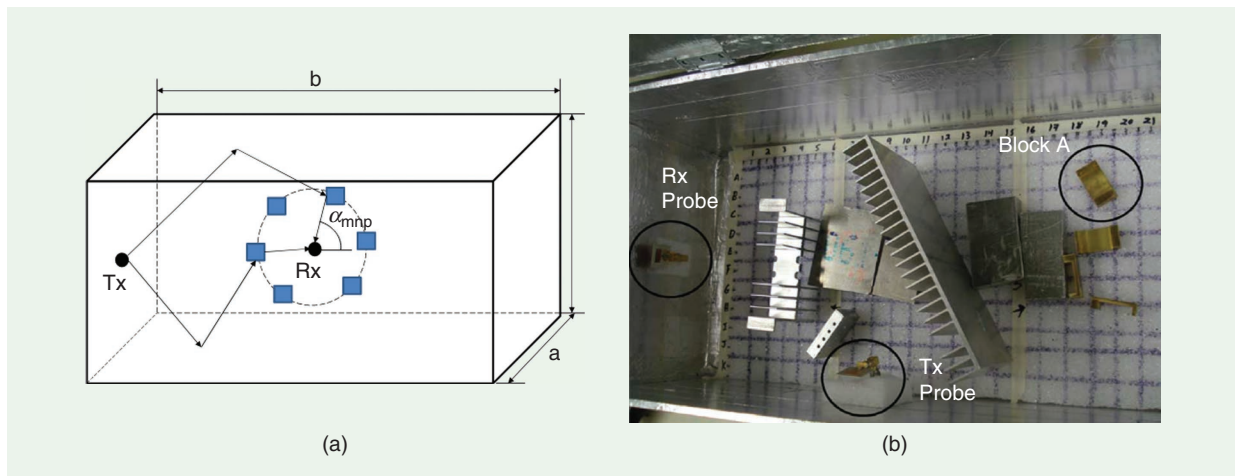


Figure 5. (a) An overmoded cavity excited with a transmit probe Tx and a receive probe Rx and loaded with conductive scatterers (blue squares). (b) The cavity with a multitude of metal scattering surfaces in [29].

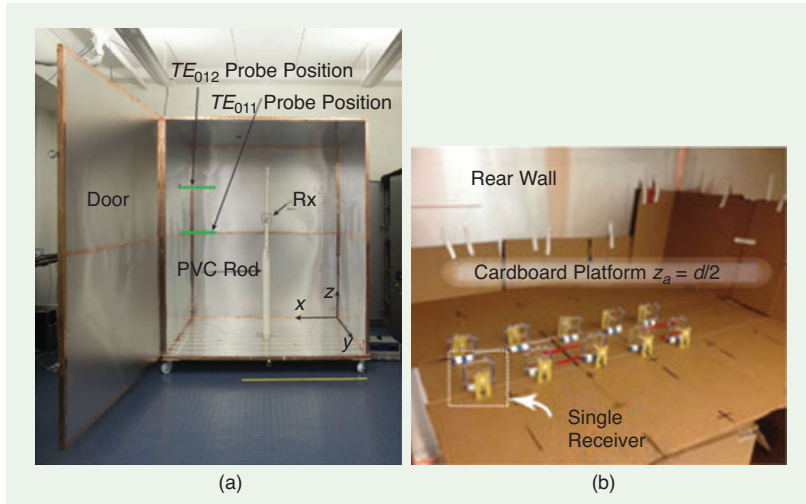


Figure 6. (a) An experimental aluminum cavity, with a yellow yardstick for scale. (b) The 10 WPT Rx's attached to half-wave rectifiers that power white LEDs in [31]. PVC: polyvinyl chloride.

(PTE) in CRs. This was accomplished by developing a bandpass filter (BPF)-equivalent model and utilizing network analysis. Experimental results were shown to closely match the developed BPF model and analytically derived maximum achievable power transfer

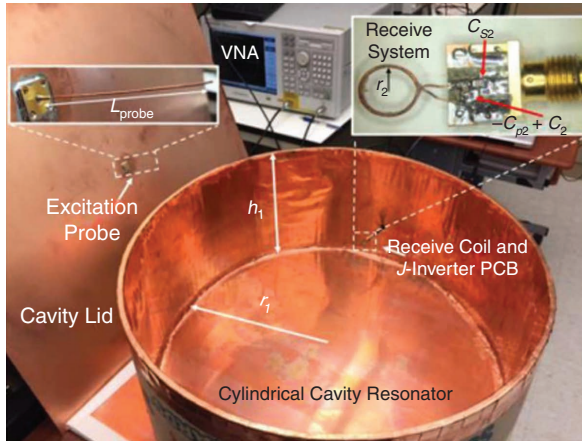


Figure 7. The cylindrical cavity and receive resonator of the WPT system in [32]. VNA: vector network analyzer; PCB: printed circuit board.

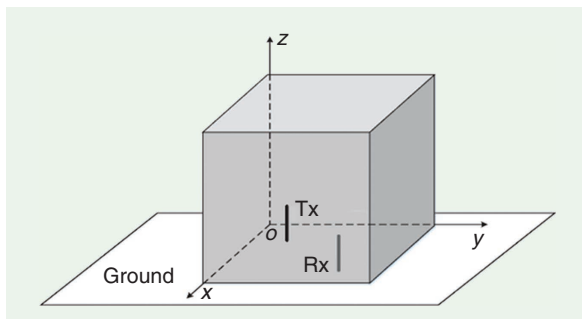


Figure 8. The simulation mode for the loss analysis in [33].

efficiencies. A peak PTE of 33% to a 5-mm-diameter receive coil (Figure 7) was achieved, which is four times larger than the PTE achieved in a system with no optimal intermediate modulation (IM) network [32].

In 2016, Liu et al. used EM field theory to analyze the EM field distribution inside a closed cavity and to explore the interaction between the EM field and the cavity structure. Based on the outcomes of EM field analysis, a power loss model (Figure 8) for wireless power transmission within a closed cavity is established, and the factors influencing the power loss are quantitatively assessed, including the return loss, power loss resulting from the nonideal material of the cavity wall, and heat loss due to

antenna resistance. This research shows that the return loss is higher than the antenna resistance loss (copper), which is higher than the cavity loss (aluminum) when it is near the resonance frequency (424 MHz) [33].

In 2016, Shahmohammadi et al. introduced and validated a generalized circuit model for cavity resonance wireless power systems [34]. The model allowed for rapid design iteration and contributed to a better understanding of how to optimize system performance. A simple impedance matching network was designed using the developed circuit model for wireless power transmission to 52 LED Rx's (Figure 9) simultaneously, with an efficiency of about 30%. This was achieved by combining two cavity models (TE_{011} and TE_{012}). The experiment showed that the proposed equivalent circuit model can be a useful tool to

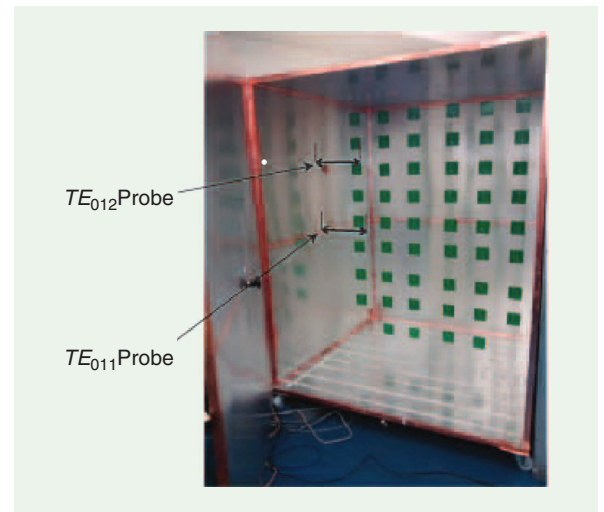


Figure 9. The aluminum chamber with 52 LED circuits in [34].

analyze cavity mode-enabled WPT systems.

In 2017, Sasatani et al. introduced a room-scale resonator with a “pole-independent” (PI) mode (Figure 10), which operates without a pole. The surface current and magnetic field distributions of the PI and pole-dependent (PD) modes were analyzed by finite-element simulation, demonstrating how the PI mode can override the inefficient regions of the PD mode, thereby improving the overall efficiency through dual-mode operation. Simulation results show that more than 66% power transmission efficiency can be achieved anywhere in the room with dual-mode operation [35].

At the same time, Wang et al. proposed a reconfigurable scheme that includes an emitting element and multiple parasitic elements. The terminal states of the parasitic elements can be switched between open and short circuits to affect the distribution of EM fields, enabling reconfigurable WPT. Experiments were conducted within a 1-m³ cube (Figure 11) using nine parasitic elements and one transmitting element. By adjusting the end states of the parasitic elements, the PTE of the Rxs at different locations was measured. The experimental results showed that a PTE of more than 70% could be achieved at 425 MHz by adjusting the states of the nine parasitic elements at any location within a 0.4 × 0.6-m area inside the cube [36], [37], [38].

In 2017, Takano et al. investigated CR WPT in an enclosed space with scatterers, which is independent of the location of Tx and Rx. CR WPT is characterized by high transmission efficiency and adaptability. The maximum transmission efficiency was investigated when the Rx end was placed at both line-of-sight (LOS) and non-LOS (NLOS) positions, and it was found that the CR WPT method can achieve a transmission efficiency of more than 30% even at the NLOS position. An experimental system (Figure 12) was constructed to drive a batteryless sensor using the CR WPT technique, and the experimental results show that the CR WPT technique can achieve both WPT and communication, but the RF–dc transmission efficiency is low and needs to be further optimized [39].

In 2017, Tamura et al. designed a closed-space model consisting of a metal grid, a transmitting probe, a receiving probe, and a frequency-selective surface (FSS) frame (Figure 13). The transmission characteristics of the FSS in the enclosed space were verified by simulation and experiment. The experimental results

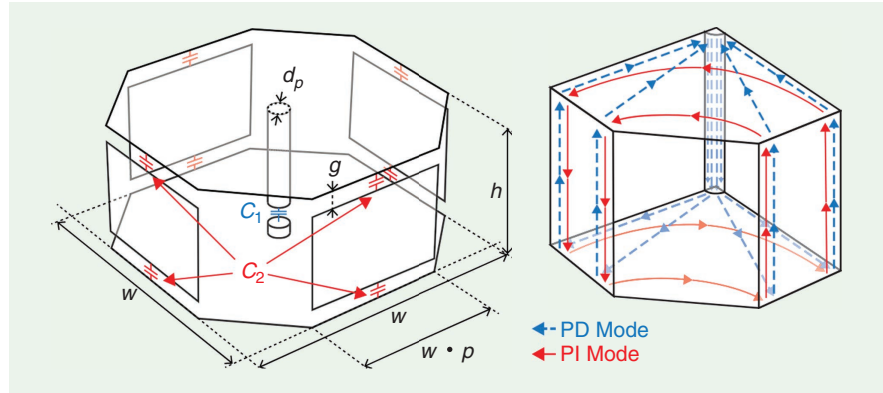


Figure 10. The geometry of the proposed resonator in [35]: the (a) full structure and (b) one unit structure. PD: pole dependent; PI: pole independent.

showed that power could be efficiently transmitted in a confined space using an FSS and that it was possible to transmit information to outside sensors while keeping the power confined. This indicated the high efficiency

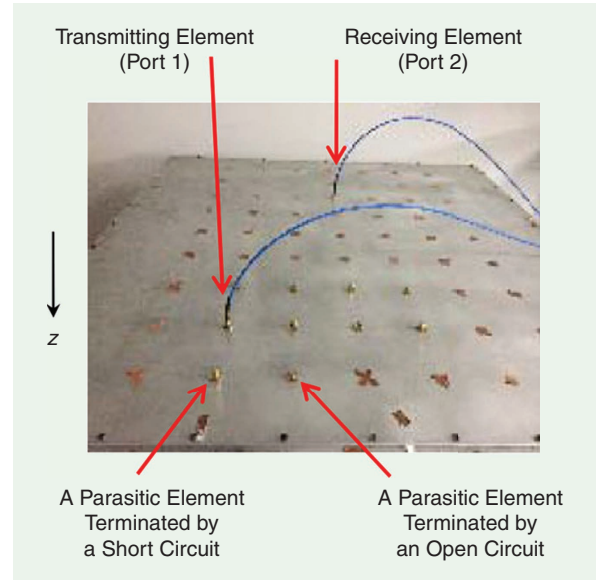


Figure 11. The experimental setup in [38].

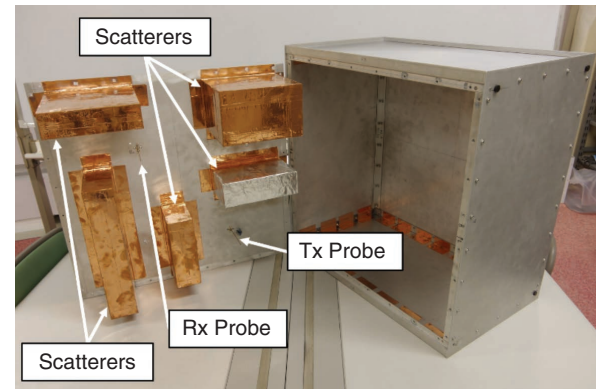


Figure 12. The experimental model with scatterers in [39].

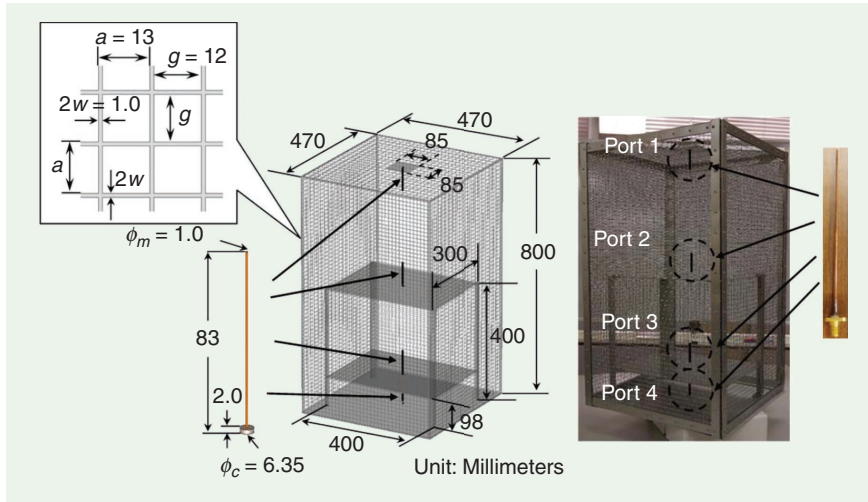


Figure 13. The metal mesh cavity in [40].

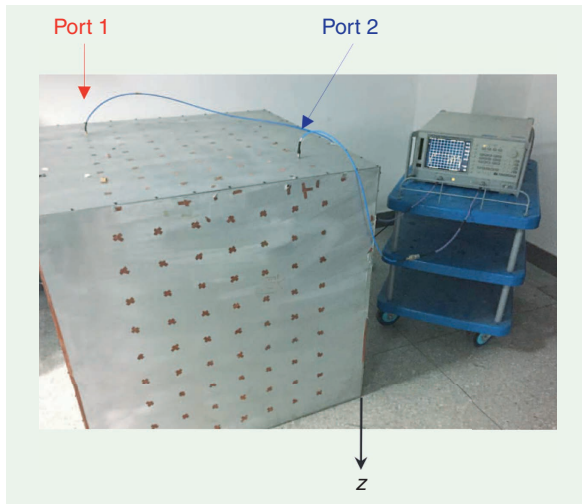


Figure 14. The experimental setup in [41].

of the power transmission and the stable transmission of information at a specific frequency [40].

In 2019, Wang et al. investigated the characterization of a wireless channel in a completely closed environment, analyzing the wireless channel inside a rectangular cavity in detail by means of semianalytical modal analysis and verifying the theoretical results experimentally. A cubic cavity (Figure 14) was analyzed, and excellent wireless channels associated with specific cavity modes (e.g., TM_{220} , TM_{310} , and

TM_{311} modes) were predicted. The experiments were carried out in a cubic metal box with sides of about 1 m. The walls of the box were made of aluminum plates, and transmitting and receiving probes were placed inside the box. The scattering parameter S_{21} was measured at various frequencies using a network analyzer, which showed that the amplitude of S_{21} was close to one near the predicted resonant frequency of the modes and that the phase varied linearly with the frequency [41].

In 2019, Furusu and Tamura focused on the design methodology of capacitive planar power Tx and differential power Rx designed for CR WPT and their performance verification. Experiments were conducted in a metal mesh cavity (Figure 15) to verify the performance of the Tx and Rx. Measurement results showed that the RF-to-dc efficiency of the proposed differential Rx was

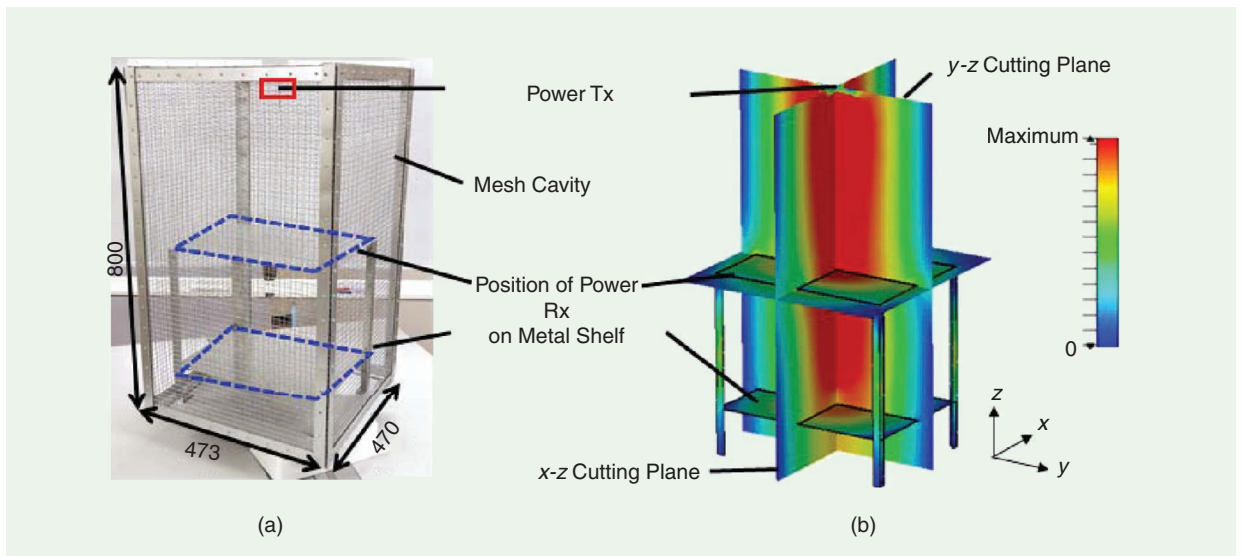


Figure 15. The (a) experimental model and (b) electric field distribution in [42].

higher than that of a conventional monopole probe at multiple receiving positions [42].

In 2020, Nimura and Tamura discussed and verified the effectiveness of a method to improve the transmission efficiency of CR WPT (Figure 16) by utilizing variable-reflection probes (RPs) terminated with a variable-reactance circuit. The effect of different RP insertion positions and numbers on the maximum value of transmission efficiency (η) was experimentally investigated.

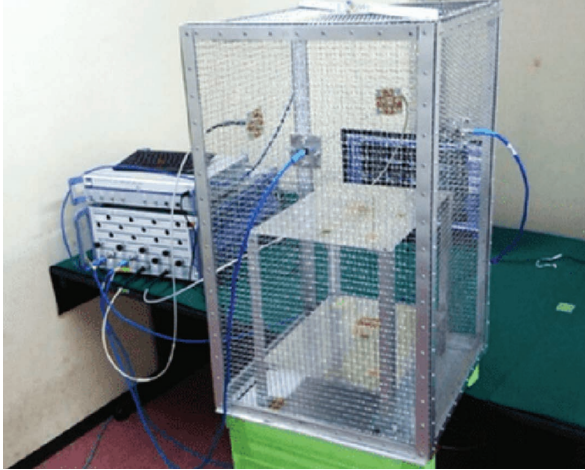


Figure 16. The experimental setup in [43].

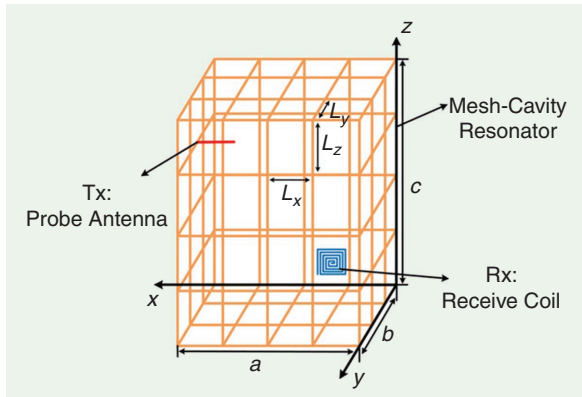


Figure 17. The 3D CR WPT system based on a mesh-cavity resonator in [44].

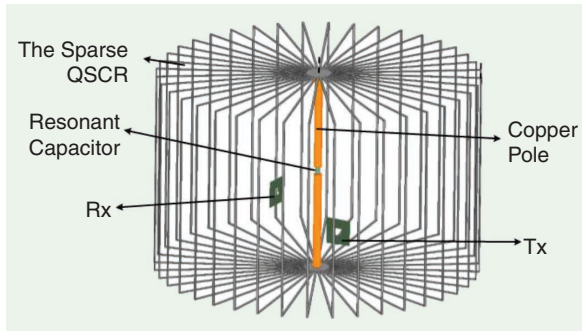


Figure 18. The CR WPT system based on a sparse quasi-static (QS) CR in [45].

The results showed that η can be improved at all receiving positions by finding RP insertion positions that yield a high η at each receiving position and inserting multiple RPs at these positions [43].

In 2020, Yue et al. presented the design and validation of a 3D WPT system based on a mesh-cavity resonator (Figure 17), which consists of a metallic grid with EM resonance modes similar to those of an all-metal closed cavity. This mesh structure reduced the cost and simplified the realization while maintaining the ability to communicate with the outside environment. The mesh-cavity resonator was simulated by full-wave EM simulation software and compared with an all-metal closed cavity. The results showed that the two cavity types generate similar magnetic field distributions. Simulations showed that wireless power can be delivered with over 70% efficiency to the Rx in a mesh cavity [44].

In 2021, Yue et al. introduced a nonradiative CR WPT system (Figure 18) based on a sparse quasi-static (QS) CR to solve the communication signal blocking problem in conventional CRs. The sparse QSCR

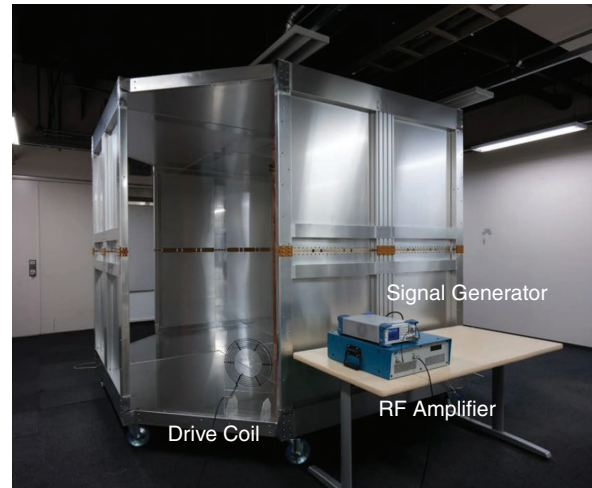


Figure 19. The constructed room-scale resonator in [46].

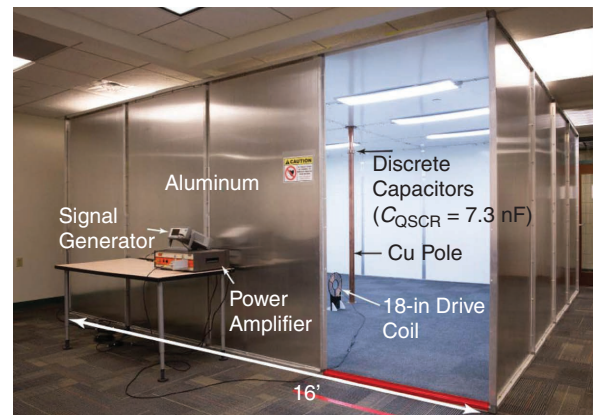


Figure 20. The room-sized experimental QSCR in [47].

was constructed of metal strips, which were designed according to the current path and LC resonance theory of the CR WPT system. A low-frequency QS EM field is excitedly generated by connecting a copper post

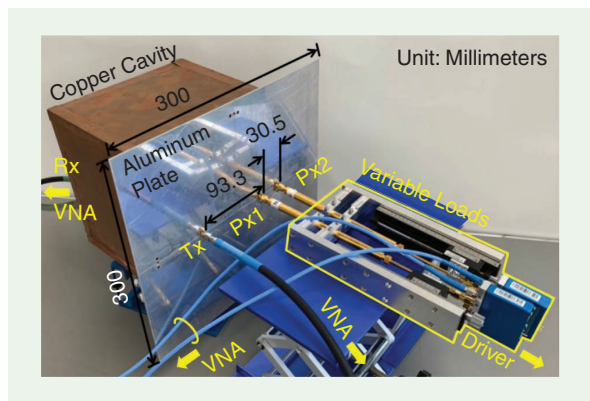


Figure 21. The experiment setup in [48].

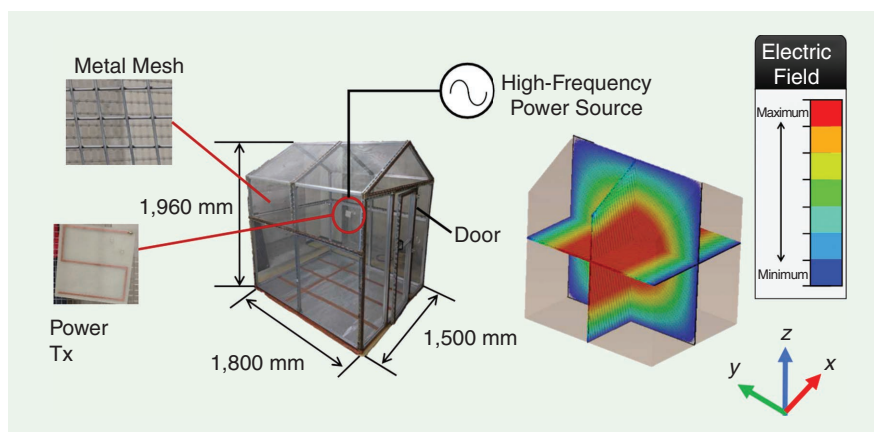


Figure 22. The gable-type cavity and analysis of the electric field distribution in [49].

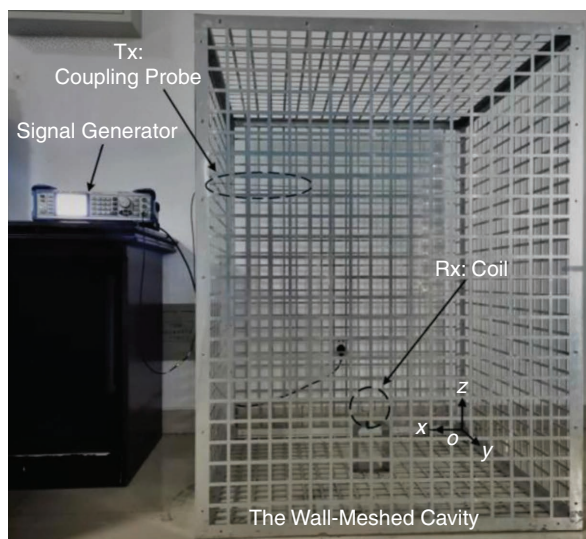


Figure 23. The wall-meshed CR-based CR WPT experimental system in [50].

with a resonant capacitor at the center of the metal strip. This EM field transmits power to the Rx in the cavity through magnetic resonance coupling for efficient wireless power transmission. Simulation results showed that the system was capable of transmitting power with an efficiency of 40% to 70% at almost any location within the cavity [45].

In 2021, Sasatani et al. designed a room-scale multimode resonator that realized multiple resonant modes with unique magnetic field distribution patterns through a combination of conductive surfaces and lumped capacitors. A $3 \times 3 \times 2$ -m test room was constructed (Figure 19), and the performance of the multimode resonator in room-scale WPT was experimentally verified. The results showed that the power transmission efficiency exceeds 37.1% in the room as a whole, exceeding 50% in most areas, with a maximum efficiency that exceeds 70%. The EM field exposure levels were evaluated to ensure

the WPT system's compliance with international safety standards. Simulation analysis demonstrated that the system could safely provide power exceeding 50 W before exceeding human exposure safety guidelines [46].

In 2022, Sasatani et al. proposed a circuit model inspired by magnetically coupled resonators, including the drive coil, the QSCR itself, and the Rx. A QSCR is a hollow conductive structure, typically consisting of a rectangular metal shell, a central metal pole, and a lumped capacitor inserted inside the pole. Current flowed through the pole–ceiling–wall–floor–pole loop, generating a 3D magnetic field. A near-field standing-wave magnetic field inside the room was generated by controlling the current flow for WPT. A $4.9 \times 4.9 \times 2.3$ -m rectangular QSCR (Figure 20) was used for experiments to measure the circuit parameters and validate the theoretical model. The effects of different geometrical parameters on the performance of the QSCR were analyzed, and a method to optimize the radius of the central pole to maximize the quality factor was proposed [47].

In 2022, Murata et al. utilized in situ S-parameter estimation among the antennas to predict the reflection characteristics of parasitic antennas (Px) under arbitrary loading conditions by observing the variation of the reflected signal at the Tx. The process of estimating the S-parameters was simplified by adding a single-pole, double-throw switch to the conventional

system. To avoid falling into a local minimum, a parallel minimum reflection search was proposed to find the global minimum by preparing multiple random initial phase pairs and applying the gradient method. Experiments were conducted in a 200-mm metal cavity (Figure 21) at 2.4 GHz. Experimental results showed that the new method improved the transmission efficiency by up to 14.4 times compared to using a single Px. The new method effectively eliminated the blind spots in the conventional method through the spatial diversity effect of dual Pxs [48].

In 2022, Akai et al. designed a cavity featuring a gable roof structure (Figure 22), measuring 1,500 mm in width, 1,800 mm in depth, and 1,960 mm in height. The cavity consisted of a metal grid with a 12.5-mm mesh spacing and a 1-mm mesh wire diameter, which was capable of reflecting EM waves while allowing wireless communication. The result of supplying power to 10 receiving modules simultaneously was demonstrated experimentally, with each module receiving at least 8.7 mW of power, which is sufficient to operate wireless sensor networks when the input power is 2 W. The transmission efficiency depended on the strength of the electric field inside the cavity and could reach up to 31.7%, which proves the feasibility of CR WPT technology to power multiple sensors simultaneously. The performance of wired and wireless communication devices and the leakage electric field strength around the cavity were tested during WPT. The results showed no packet loss and unaffected communication delays at input powers up to 5 W, validating the system's ability to supply power while keeping communication open. The maximum leakage electric field strength was 0.95 V/m, which was much lower than the safety standard value of 27.5 V/m, ensuring the safety of the system [49].

In 2022, Yue et al. designed a CR (Figure 23) consisting of a mesh metal wall, which was analytically designed using known theoretical equations without the need for time-consuming numerical optimization. The effectiveness of the design was verified by EM simulations and experiments, and the mesh metal wall was able to limit the resonance of EM waves inside the cavity while allowing wireless communication signals to penetrate. Experimental results showed that the system achieved efficient WPT inside the cavity, with

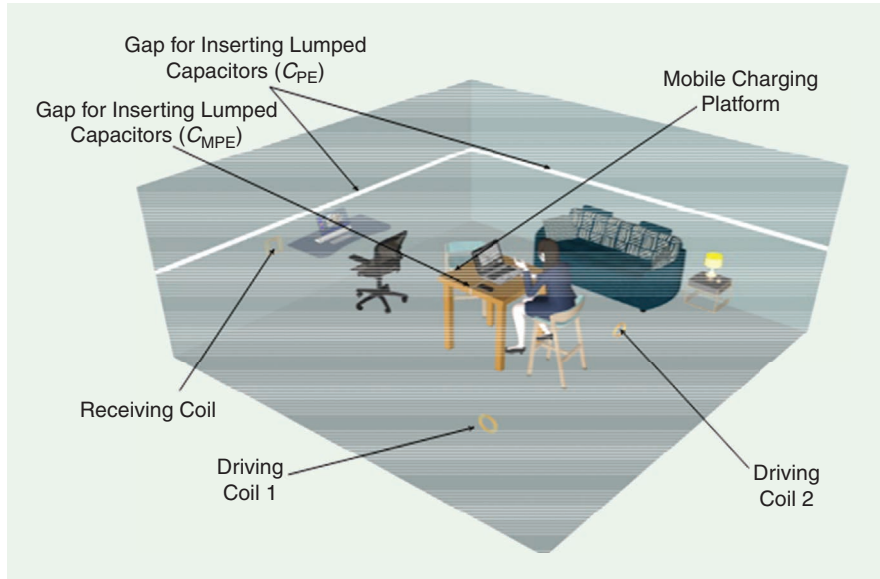


Figure 24. The space-utilizing multimode QS cavity resonance system in [51].

a maximum transmission efficiency of more than 85% and transmission efficiency of more than 60% over about 70% of the cavity area. For wireless communication signals (e.g., 4G signals at 2.1 GHz, Wi-Fi signals at 2.45 GHz, and 5G signals at 3.5 GHz), the attenuation through the mesh metal wall was 6, 5.1, and 3.8 dB, respectively, which proved the low attenuation characteristics [50].

In 2023, Rong et al. proposed a space-utilized multimode QS resonant cavity (SUM-QSRC) WPT method to improve space utilization by introducing a movable platform (Figure 24) to enhance the local magnetic field. The SUM-QSRC combined two independent resonant modes: a peripheral field enhancement (PE) mode and a movable platform field enhancement (MPE) mode. The two modes did not interfere with each other during operation. Characteristic frequency analysis and magnetic field distribution verification of the SUM-QSRC system was carried out by finite-element software, which proved the independence and noninterference of the PE mode and the MPE mode. The results showed that, with the assistance

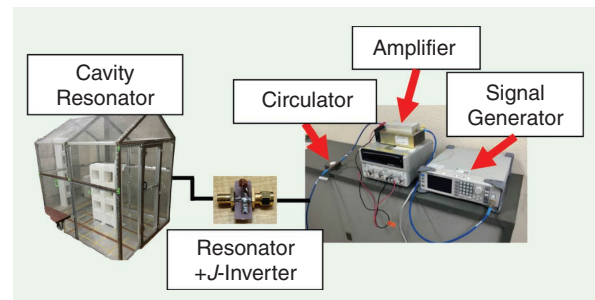


Figure 25. The experiment setup in [52].

TABLE 1. A comparison of WPT solutions for closed spaces.								
Reference	Cavity Type	Frequency (GHz)	Volume (cm³)	Method	Field Distribution	Remarks (▲ Limitations)	Tx and Rx	Efficiency
[26]	Rectangular cavity with four sidewall AMC surfaces	0.8–1.3	19.6 × 10 × 10	AMC surfaces	Uniform	• Broadband • Metamaterial (AMC) • Compact size	Monopoles	0.6-dB insertion loss
[27]	Overmoded rectangular cavity	2.2	50 × 25 × 20	Multiple excited modes and narrow-band frequency modulation	Uniform	• Scalable • Power multiple devices simultaneously	Patch-type probes	
[28]	Overmoded rectangular cavity	10	49.7 × 24.7 × 18.7	Mechanically and frequency-modulated stirring	Uniform	• Scalable • Reverberation chamber • Power multiple devices simultaneously	Custom-made patch-type microstrip probes	
[29]	Overmoded rectangular cavity	10	50 × 28 × 18	Placing random scattering objects	Uniform	• Geometrical propagation model allows statistical prediction • Power multiple devices simultaneously	Patch-type probes	
[30]	Resonant rectangular cavity		152 × 142 × 183	CMT	Nonuniform and fixed	• Derive the coupling coefficient • Predict the WPT efficiency ▲ Unintuitive	Short dipoles	Over 60%
[31]	Multimode resonant rectangular cavity	0.132 and 0.193	152 × 142 × 183	Stimulating natural EM resonant modes with low-level EM fields	Uniform	• 3D charging • Power multiple devices simultaneously ▲ Does not consider any Rx optimization	Square-shaped coil and linear probe	Most area exceeds 50%
[32]	Cylindrical resonant cavity	0.375	$\pi \times 30^2 \times 25.4$	BPF theory and microwave network systems analysis	Nonuniform	• A simpler alternative to CMT • IM network ▲ Not universal	Circular coil and probe	Peak PTE 33.88%
[33]	Resonant rectangular cavity	0.424	100 × 100 × 100	EM field theory	Nonuniform	• Power losses analyzed	Probes	Over 80%
[34]	Resonant rectangular cavity	0.132 and 0.193	152 × 142 × 183	Combining two cavity modes	Nonuniform	• IM network • Intuitive circuit model • Power multiple devices simultaneously	Square-shaped coil and linear probe	30%
[35]	Multimode QS cavity	0.00132 and 0.00133	54	CMT and PI mode	Nonuniform	• Room-scale QSCR • PI • Eliminate the null zone of power delivery ▲ Corners of the cavity are removed	Square helix coil	50% (PI mode) 60% (dual mode)

(Continued)

TABLE 1. A comparison of WPT solutions for closed spaces. (Continued)

Reference	Cavity Type	Frequency (GHz)	Volume (cm ³)	Method	Field Distribution	Remarks (▲ Limitations)	Tx and Rx	Efficiency
[38]	Resonant rectangular cavity	0.425	100 × 100 × 100	One transmitting element and multiple parasitic elements	Nonuniform	• Reconfigurable • Low implementation cost ▲ Based upon trial and error	Probes	Over 70% within a 0.4 × 0.6-m region
[39]	Resonant rectangular cavity and metal mesh	0.357	49.6 × 47.7 × 29.1	Time-division power feeding and CR WPT with scatterers	Nonuniform	• NLOS positions considered • Possible to transfer power and communicate simultaneously ▲ Low RF–dc transmission efficiency	Helix-type probes	Over 30% in the NLOS position
[40]	Metal mesh cavity with FSSs	0.447	47 × 47 × 80	FSSs	Nonuniform	• NLOS positions considered • FSS • WPT	Capacitive probes	
[41]	Resonant rectangular cavity	0.395–0.435	98.6 × 98.6 × 98.1	Semianalytical modal analysis and cavity Green's function	Nonuniform	• Predict the characteristics of practical wireless channels	Probes	
[42]	Metal mesh cavity	0.45	47 × 47.3 × 80	Capacitive coupling of <i>J</i> -inverter	Nonuniform	• Higher RF-2020 efficiency achieved than conventional monopoles	Capacitive probes consist of one or two inverted L-shaped open stubs	
[43]	Metal mesh cavity	0.45	47 × 47.3 × 80	Utilizing RPs terminated with a variable-reactance circuit	Nonuniform	• Transfer power to all receiving positions and achieve higher efficiencies	Probes	
[44]	Metal mesh cavity	0.195	152 × 142 × 183	EM shielding effect and standing-wave formation theory	Nonuniform	• Solve problems, such as high cost, difficult implementation, and inability to communicate with outside	Probe and square receive coil	Over 70% within the 4-m ³ meshed cavity
[45]	Sparse QSRC	0.0013	$\pi \times 80^2 \times 50$	Current path of the CR WPT system and the LC resonance theory	Nonuniform	• Possible to communicate with the outside world ▲ Need for a copper pole connected to the metallic strips in the center	Transceiver coils	40% to 70% at almost any location within the cavity

(Continued)

TABLE 1. A comparison of WPT solutions for closed spaces. (Continued)

Reference	Cavity Type	Frequency (GHz)	Volume (cm ³)	Method	Field Distribution	Remarks (▲ Limitations)	Tx and Rx	Efficiency
[46]	Multimode room-scale QSRC	0.00134	300 × 300 × 200	Magneto-QS fields, CMT, and cutting the corners of the room	Nonuniform	• Safely deliver over 50 W of power • Central pole can be omitted • Flexibility in system dimensions • Low interference caused by external objects ▲ Rx coil needs to be orientated orthogonally with the ambient magnetic field	Transceiver coils	Over 37.1% throughout the room
[47]	Room-scale QSRC	0.00132	490 × 490 × 230	CMT	Nonuniform	• Effective circuit model that can accommodate variations in QSCR room geometry as well as Rx size, orientation, and position	Transceiver coils	Over 50% in most of the volume
[48]	Resonant rectangular cavity	2.4	20 × 20 × 20	Closed-form S-parameter estimation and parallel minimum reflection searching	Uniform	• Implicit CR WPT • Improved transfer efficiency and spatial uniformity	Monopoles	50% at most measured points
[49]	Gable-type metal mesh cavity	0.12	180 × 150 × 196	CR WPT and L-type matching circuit	Nonuniform	• Simultaneous power supply to 10 power-receiving modules • Wired and wireless communication available • Safety considered	Probe with an S-shaped open stub and spiral-type differential power Rx	15.9% to 31.7%
[50]	Wall-meshed resonant cavity	0.278	91.6 × 91.6 × 120.4	CMT and EM shielding theory	Nonuniform	• Support wireless communications with outside simultaneously • More uniform PTE distribution • Safety considered	Probe and coil Rx	Over 60% within 70% of the areas inside the cavity (maximum: over 85%)
[51]	Multimode QSRC	0.647 and 1.805	200 × 200 × 180	PE mode and MPE mode	Nonuniform	• WPT without dead angles • Improve space coherence	Coils	Over 40% at any position (maximum: 90%)
[52]	Gable-type metal mesh cavity	0.12	180 × 150 × 196	Wideband BPF design theory and J-inverter	Nonuniform	• Broadband WPT	Patch with the fabricated resonators and J-inverters	20%, with 3× the conventional bandwidth

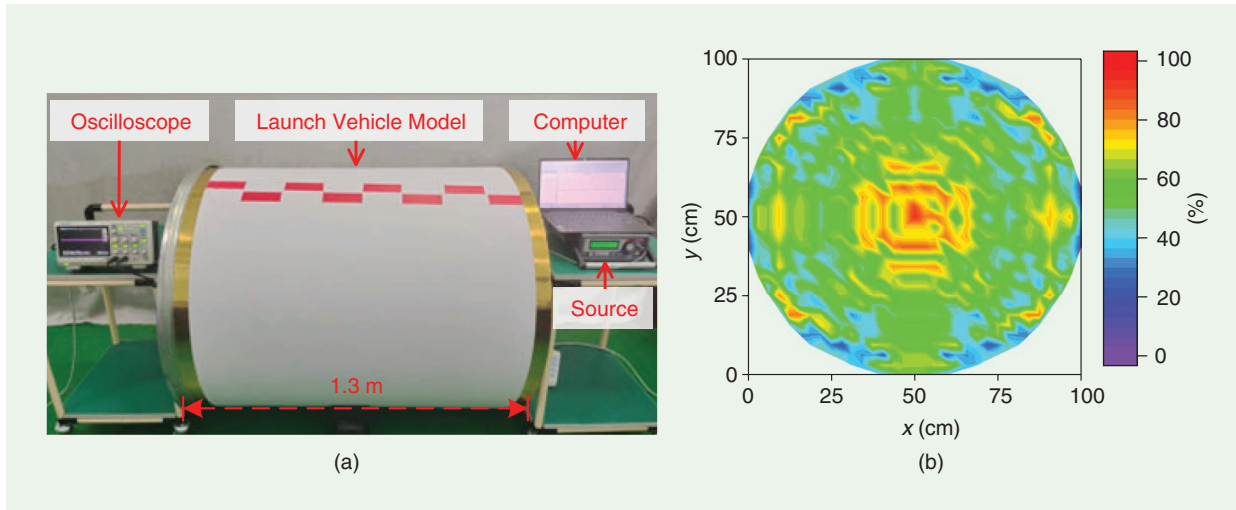


Figure 26. The (a) launch vehicle model and (b) power (microwave-to-microwave) transmission efficiency.

of the movable platform, the PTE could reach more than 40% in the space as a whole and up to 90% in some areas [51].

In 2024, Tamura et al. proposed a filter design theory-based design method for CR WPT systems to realize broadband transmission frequency and efficient WPT. A *J*-inverter (Figure 25), external and internal additional resonators, and a third-order BPF were introduced and configured for broadband matching. The bandwidth was broadened from about 0.6 to

2 MHz in the 117–122-MHz band, which was about three times the previous system [52].

Most of the WPT systems for closed spaces discussed above are based on the CMT. Most of them are rectangular cavities, while a few of them involve cylindrical cavities or hexagonal cavities. Some of the researchers use fully enclosed metal cavities, while the others employ metal grids. While the majority of the earlier literature is devoted to the study of efficient and uniform wireless power supplies to realize

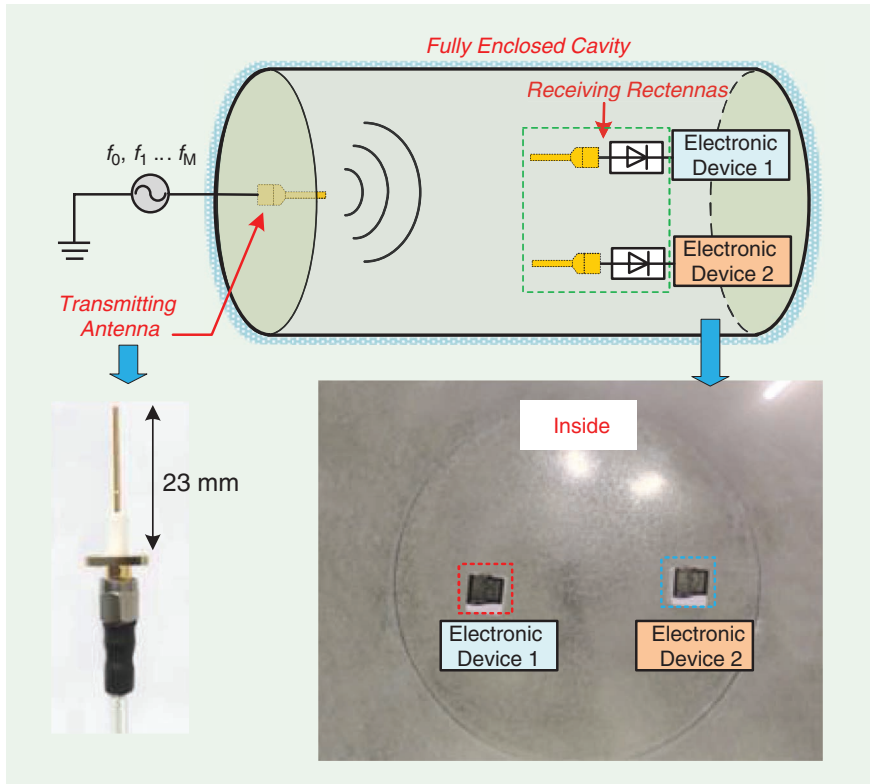


Figure 27. The system configuration and wirelessly powering dual sensors.

WPT at arbitrary positions, some researchers are also committed to realizing power supply to multiple Rx simultaneously and to wireless communication during power transmission. Most researchers use monopole probes as transmitting and receiving ends in experiments. A significant focus of the research is optimizing the design of the Tx and Rx for better WPT performance. Importantly, the safety levels of microwave leakage and input power have also been studied. A comparison of each solution's performance, strengths, and weaknesses is given in Table 1.

Recently, Liu's team proposed a microwave wireless power transmission technology inside a closed cavity based on a novel frequency controlling technique that can realize controllable high-efficiency

wireless charging to multidirectional sensors in an electrically large cavity [25]. Figure 26(a) shows an enclosed space WPT system that takes a launch vehicle model with a length of 1.3 m as an example. The cavity is an electrically large cavity, and its size is much larger than the working wavelength. The cylinder features two simple monopole antennas positioned on its upper and lower surfaces, which are utilized for measuring the S-parameters between them. By changing the working frequency, full coverage of the cavity can be achieved, and the maximum microwave-to-microwave power transmission efficiency is 96.6%, as evident in Figure 26(b). The efficiency is calculated through $\eta_{\text{RF-RF}} = 10^{\frac{|S_{21}|}{10}} \times 100\%$, where $|S_{21}|$ represents the voltage transmission coefficient in decibels.

The cavity has thousands of EM modes with respect to a frequency band, which means that the frequency gap between EM modes is very narrow. Each EM mode inside the cavity corresponds to an operating frequency. Therefore, the field distribution inside the cavity is very sensitive to changes in frequency, and a small change in frequency will change the EM mode of the cavity, resulting in a completely different EM field distribution. The average frequency interval between adjacent EM modes of the cavity is about 180 kHz. Taking the Wi-Fi frequency band as a demonstration, multiple operational states of dual Rx's can be achieved through frequency control, enabling the simultaneous powering of either a single sensor or of dual sensors, as depicted in Figure 27. The success of this experiment shows the potential in the application of wireless power supply for sensors in closed spaces, such as aerospace vehicles.

Liu's group performed a further experiment to study a frequency-controlled WPT in a cavity. They demonstrated another approach to WPT in closed cavities using a metal grid cubic cage, as in Figure 28. The cage for WPT at the S band is $1.3 \times 1.3 \times 2$ m, which is also an electrically large cavity. A simple monopole transmitting antenna is mounted at the top of the cage, and 144 rectifying antennas are configured in a square arrangement at the bottom of the cage, forming a 12×12 array. Each rectenna has an LED load to show whether it collects power. Then, the authors trained the WPT system to obtain a mapping relationship from a specific operating frequency to light an LED at a specific location. Therefore, by adjusting the working frequency, the LEDs can be lit at specific positions to form a specific track, such as a circle or a rectangle. This shows that the wave propagation can be manipulated to select the receiving targets of WPT in a closed cavity by controlling input frequencies.

Future Applications

There are many situations where closed or semiclosed cavities can be applied (e.g., a section of a launch vehicle, the cockpit of an airplane, or the cab of a car). Furthermore, once we understand the WPT mechanism inside a cavity, we can enhance the cavity boundary to form more fully closed cavities and, thus, enhance WPT performance. For example, the glass of a car window leaks microwaves, breaking the integrity of the cavity. However, indium–tin oxide (ITO) films can be used to cover the glass, keeping it transparent while working as a conductor. Then, the car cab becomes a good cavity, and electronic devices can be wirelessly powered

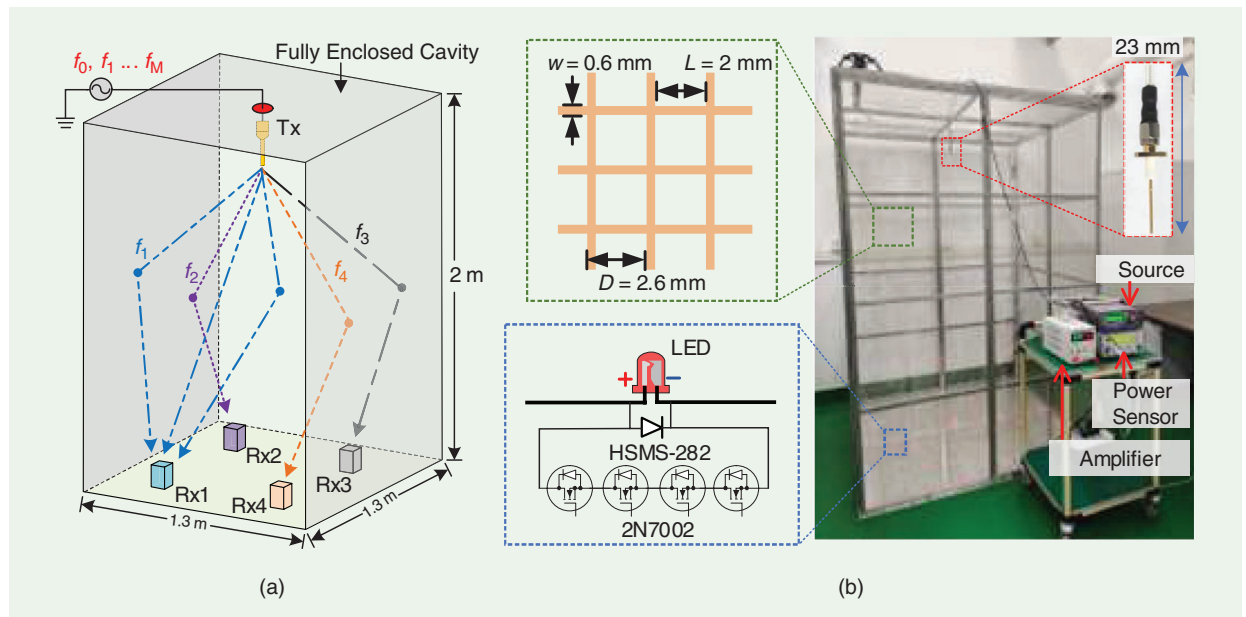


Figure 28. The (a) WPT demonstration in (b) a metal grid cubic cage.

using WPT. Some specially treated ITO films have a high temperature tolerance range, so it is safe to use ITO films instead of glass, even if the car is exposed to the sun. The use of ITO films in cars would not significantly raise manufacturing costs, due to their thin and lightweight properties. Additionally, mass production can decrease unit costs through optimized production processes and improved efficiency.

A smart home is a convenient setup where home appliances and electronic devices can be controlled automatically and remotely with an Internet connection. Smart home devices allow the user to control their functions remotely, such as remotely adjusting the home's temperature, lighting, and humidity. All appliances would benefit from a flexible power supply for convenience. Room-scale magneto-QS wireless power transmission using a cavity-based multimode resonator may someday supply power to home appliances [46]. Home appliances like fans, lamps, and phones can also be powered wirelessly. Once the roof, walls, and floor of a room are built to reflect microwaves at certain frequencies (e.g., with metasurfaces), and thus form a closed cavity at room scale, home appliances and electronic devices may one day be powered by microwave WPT.

Summary

Wireless power transmission technology in closed cavities holds great promise across multiple fields. Due to weak microwave leakage from a cavity, small transmitting and receiving antennas can achieve high transmission efficiency. Moreover, wireless power transmission to multiple objects can be controlled by tuning the operating frequency.

WPT in cavities can be applied to industrial automation and consumer electronics. In industrial automation, this technology can provide a continuous power supply to sensors, robots, or other equipment within closed spaces, enhancing production efficiency and safety. In consumer electronics, WPT offers convenient and efficient charging solutions for smartphones, wearable devices, and more, freeing users from the constraints of power cords and significantly improving the user experience. With continuous advancements in technology and concomitant cost reductions, WPT in closed environments is expected to see widespread application in more fields, heralding a revolution in energy supply methods.

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